

Algorithms

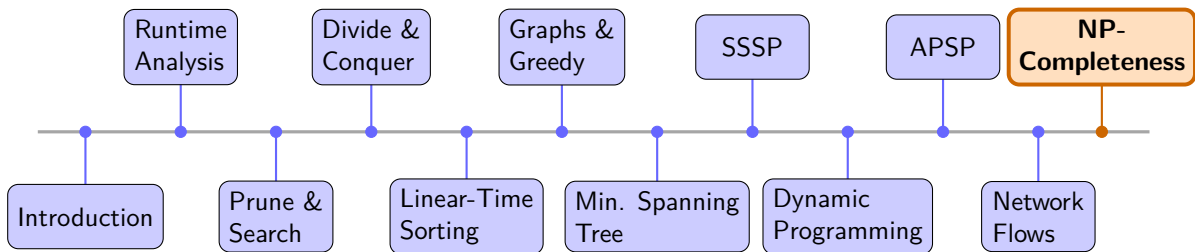
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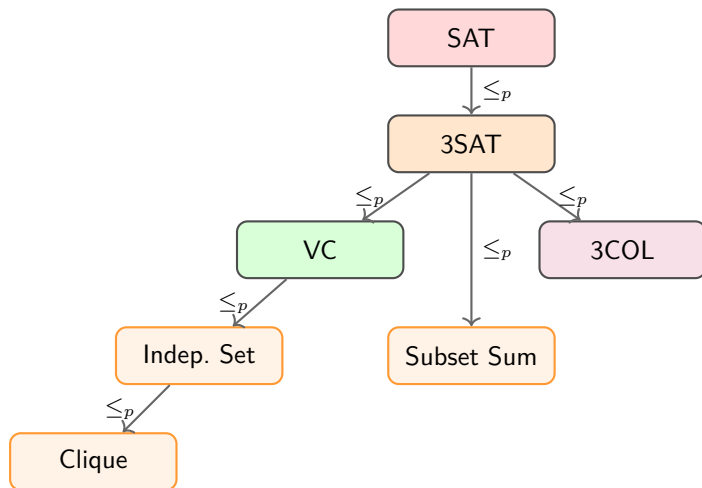
April 27, 2026



Recap & What's Next



Where We Are: The NPC Chain So Far



Subset Sum

Definition

Given a multiset of positive integers $S = \{s_1, s_2, \dots, s_n\}$ and a target t .

Subset Sum: Does there exist a subset $S' \subseteq S$ such that $\sum_{s \in S'} s = t$?

Example: $S = \{3, 1, 4, 2, 6\}$, $t = 9$.

Try $S' = \{3, 4, 2\}$: $3 + 4 + 2 = 9$. ✓

Example: $S = \{3, 7, 11, 2\}$, $t = 8$.

Possible sums: 3, 7, 11, 2, 10, 14, 18, 5?, ...

No subset sums to 8. ✗

Certificate for Subset Sum $\in$ NP: The subset S' . Verify $\sum_{s \in S'} s = t$ in $O(n)$. ✓



Subset Sum Example: Trying All Subsets

$S = \{1, 3, 5, 7\}$, $t = 8$.

Subset	Sum	= 8?
$\{1\}$	1	✗
$\{3\}$	3	✗
$\{5\}$	5	✗
$\{7\}$	7	✗
$\{1, 3\}$	4	✗
$\{1, 5\}$	6	✗
$\{1, 7\}$	8	✓

$S' = \{1, 7\}$ works. ✓

But the brute-force approach checks 2^n subsets—exponential in n .

This is why Subset Sum is not obviously in P. The question is whether a smarter algorithm exists.



Reduction: $3SAT \leq_p \text{Subset Sum}$

Input: 3-CNF formula φ with n variables $x_1, \dots, x_n$ and m clauses $C_1, \dots, C_m$.

High-level idea

Encode the formula as a Subset Sum instance using carefully chosen integers. Each integer represents a choice (variable true/false, clause satisfied). The target t is chosen so that a valid subset exists $\iff \varphi$ is satisfiable.

Construction. Use integers with $n + m$ decimal digits.

- For each variable x_i : two integers v_i and $\bar{v}_i$.
- For each clause C_j : two *slack* integers d_j and e_j .

Target t has:

- Digit i (variable position): $t_i = 1$ for $i = 1, \dots, n$.
- Digit $n + j$ (clause position): $t_{n+j} = 3$ for $j = 1, \dots, m$.



$3SAT \leq_p$ Subset Sum: Integer Construction

Integers have $n + m$ digit positions.

Integer	Digit positions		Role
	i -th var	$n+j$ -th clause	
v_i	1 in position i	1 in $n+j$ if $x_i \in C_j$	Choose $x_i = T$
$\bar{v}_i$	1 in position i	1 in $n+j$ if $\neg x_i \in C_j$	Choose $x_i = F$
d_j	0 everywhere	1 in position $n+j$	Clause j slack
e_j	0 everywhere	1 in position $n+j$	Clause j slack

All other digit positions are 0. No carries occur (digits stay ≤ 9).

Target:

$$t = \underbrace{1\ 1\ \dots\ 1}_{n \text{ digits}} \underbrace{3\ 3\ \dots\ 3}_{m \text{ digits}}$$



3SAT $\leq_p$ Subset Sum: Small Example

Formula: $\varphi = (x_1 \vee \neg x_2 \vee x_3) \wedge (\neg x_1 \vee x_2 \vee x_3)$

$n = 3$ variables, $m = 2$ clauses. Digit columns: x_1, x_2, x_3, C_1, C_2 .

Integer	x_1	x_2	x_3	C_1	C_2
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v_1	1	0	0	1	0

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$3\text{SAT} \leq_p \text{Subset Sum}$: Small Example

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[illegible]

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v_3	0	0	1	1	1
$\bar{v}_3$	0	0	1	0	0

Integer	x_1	x_2	x_3	C_1	C_2
Target t	1	1	1	3	3



$3\text{SAT} \leq_p \text{Subset Sum}$: Small Example

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Integer	x_1	x_2	x_3	C_1	C_2
d_1	0	0	0	1	0
Target t	1	1	1	3	3



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Integer	x_1	x_2	x_3	C_1	C_2
d_1	0	0	0	1	0
e_1	0	0	0	1	0
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Integer	x_1	x_2	x_3	C_1	C_2
d_1	0	0	0	1	0
e_1	0	0	0	1	0
d_2	0	0	0	0	1
e_2	0	0	0	0	1
Target t	1	1	1	3	3



3SAT $\leq_p$ Subset Sum: Correctness

Claim

φ is satisfiable $\iff$ the Subset Sum instance has a solution.

($\Rightarrow$) SAT assignment $\Rightarrow$ Subset Sum solution.

- 1 For each variable x_i : pick v_i (if $x_i = T$) or $\bar{v}_i$ (if $x_i = F$). *Variable digits*: each i -th column sums to 1. ✓
- 2 For each clause C_j : the chosen literals contribute 1, 2, or 3 to column $n+j$. Add slack (d_j and/or e_j) to bring the column sum to 3. ✓

($\Leftarrow$) Subset Sum solution $\Rightarrow$ SAT assignment.

- Variable columns each sum to 1 $\Rightarrow$ exactly one of $v_i, \bar{v}_i$ chosen. This defines a truth assignment.
- Clause column $n+j$ sums to 3. Slack integers d_j, e_j contribute at most 2. The literals chosen contribute ≥ 0 . Since slack alone contributes ≤ 2 , some literal in C_j must have been chosen $\Rightarrow$ clause C_j is satisfied. ✓



Subset Sum is NP-Complete

Theorem

Subset Sum is NP-complete.

Step 1 — Subset Sum $\in$ NP.

Certificate: the chosen subset S' . Verify $\sum_{s \in S'} s = t$ in $O(n)$. ✓

Step 2 — Subset Sum is NP-hard.

We showed $3SAT \leq_p$ Subset Sum. Since 3SAT is NP-hard:

$$\forall L \in \text{NP} : L \leq_p 3SAT \leq_p \text{Subset Sum}$$

By transitivity, Subset Sum is NP-hard. ✓



Independent Set

Definition

Given an undirected graph $G = (V, E)$ and integer k .

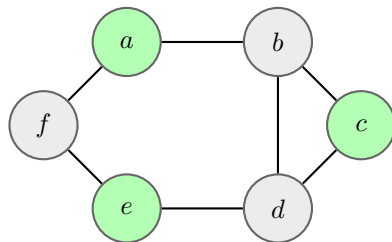
Independent Set (IS): Does there exist a set $S \subseteq V$ with $|S| \geq k$ such that *no two vertices in S are adjacent*?

Example ($k = 3$): ✓Yes

Set $S = \{a, c, e\}$: none of these share an edge.

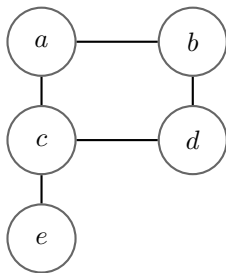
Certificate for IS $\in$ NP:

The set S itself. Check that no two vertices in S share an edge: $O(k^2)$. ✓



Independent Set Example: Checking the Condition

Let G have edges $E = \{(a, b), (a, c), (b, d), (c, d), (c, e)\}$.



Is $\{a, d, e\}$ an independent set?

Check all pairs:

- (a, d) : not an edge ✓
- (a, e) : not an edge ✓
- (d, e) : not an edge ✓

Yes: $S = \{a, d, e\}$ is an independent set of size 3. ✓

Is $\{a, b, e\}$ independent?

(a, b) is an edge. ✗



Reduction: $VC \leq_p IS$ (and $IS \leq_p VC$)

Complement Theorem

S is a vertex cover of $G = (V, E)$ of size $\leq k \iff V \setminus S$ is an independent set of size $\geq |V| - k$.

Proof. Let $\bar{S} = V \setminus S$.

($\Rightarrow$) If S is a vertex cover, every edge (u, v) has $u \in S$ or $v \in S$. So if $u, v \in \bar{S}$, the edge (u, v) cannot exist — $\bar{S}$ is independent.

($\Leftarrow$) If $\bar{S}$ is independent, no edge has *both* endpoints in $\bar{S}$. So every edge has at least one endpoint in S — S is a vertex cover. ✓

Corollary

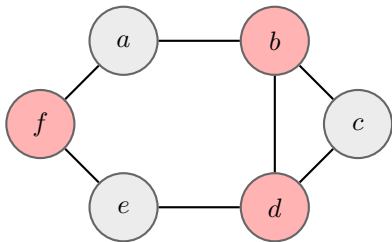
(G, k) is a Yes-instance of VC $\iff (G, |V| - k)$ is a Yes-instance of IS.

Both reductions run in $O(1)$ (same graph, just flip the parameter). ✓

VC $\leftrightarrow$ IS: Visual Example

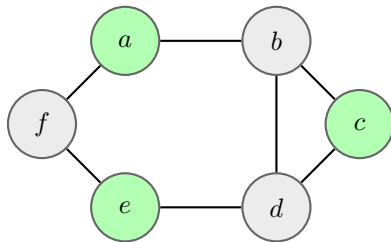
Graph G , $|V| = 6$, $k_{VC} = 3$, $k_{IS} = 6 - 3 = 3$.

Vertex Cover $S = \{b, d, f\}$



$S = \{b, d, f\}$ covers all edges

Independent Set $\bar{S} = \{a, c, e\}$



$\bar{S} = \{a, c, e\}$: no two adjacent



Clique

Definition

Given an undirected graph $G = (V, E)$ and integer k .

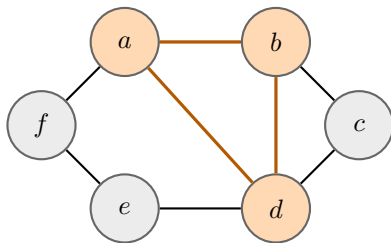
Clique: Does there exist a set $S \subseteq V$ with $|S| \geq k$ such that *every pair of vertices in S is adjacent*?

Example ($k = 3$): ✓Yes

$S = \{a, b, d\}$: all three pairs are edges.

Certificate for Clique $\in$ NP:

The set S . Verify all $\binom{k}{2}$ pairs are edges:
 $O(k^2)$. ✓



Reduction: $IS \leq_p \text{Clique}$

Complement Graph

The **complement** $\bar{G} = (V, \bar{E})$ has the same vertex set and

$$\bar{E} = \{(u, v) \mid u \neq v \text{ and } (u, v) \notin E\}.$$

Key Observation

S is an independent set in $G \iff S$ is a clique in $\bar{G}$.

Proof.

$(\Rightarrow)$ No edge between any two vertices of S in $G \Rightarrow$ every pair is an edge in $\bar{G} \Rightarrow S$ is a clique in $\bar{G}$. ✓

$(\Leftarrow)$ Every pair in S is an edge in $\bar{G} \Rightarrow$ no pair is an edge in $G \Rightarrow S$ is independent in G . ✓

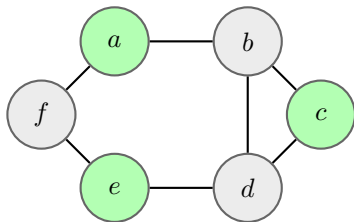
Reduction: Given (G, k) for IS, output $(\bar{G}, k)$ for Clique. Computing $\bar{G}$ takes $O(|V|^2)$. ✓



$IS \leq_p$ Clique: Visual Example

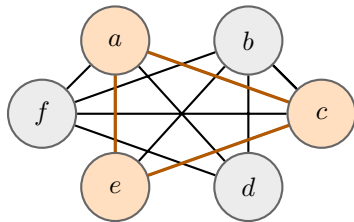
Graph G

Independent set $\{a, c, e\}$: no edges among them.

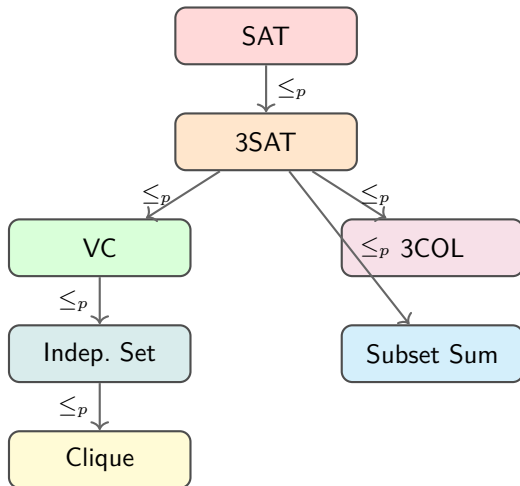


Complement $\bar{G}$

Clique $\{a, c, e\}$: all pairs now edges.



Today's Complete NPC Chain



Each arrow
 $A \leq_p B$ means:
“ B at least as
hard as A ”